PROJECT DOCUMENTATION

FMCG Manufacturing, Pricing & Profitability Analytics System

1. Project Overview

Project Name

FMCG Manufacturing & Pricing Analytics System

Domain

- FMCG (Food & Beverages)
- Manufacturing
- Pricing & Profitability
- ERP & BI Analytics

Objective

To design a **centralized analytical system** that integrates:

- Manufacturing capacity & production data
- Cost computation
- Pricing ladder (Ex-Factory → Wholesale → Retail → MRP)
- Tax logic (Pre-GST & GST)
- Profitability analysis
  for **Food and Beverage FMCG products**.

2. Business Problems Addressed

Problem	Solution
No visibility into plant utilization	Capacity utilization KPIs
Manual price calculation	Automated price ladder
Inconsistent margins	Category-based margin rules
No profitability clarity	Profit per pack & per quintal
No historical tax logic	Pre-GST vs GST handling
Difficulty in scenario analysis	Produced vs Expected vs Installed capacity

3. Data Sources

Fact Table

FACT_PRD_MANU_PROD_MASTER_MAIN_NEW_DATA

Contains:

- Daily production quantities
- Plant capacities
- Cost components
- Pack sizes
- Manufacturing dates

Dimension Tables

Table	Purpose
DM_Manufacturing_Unit	Plant details
DM_State	State mapping
DM_Material_Category	Raw material grouping
DM_Category	Product category
DM_Subcategory	Sub-category
DM_Plant_Manager	Operations ownership
DM_Product_Pack	Product packaging

4. Core Assumptions & Units

Manufacturing Units

- **1 Quintal = 100 KG**
- **1 KG = 1000 grams / ml**
- **1 Quintal = 100,000 ml**

Pricing Assumptions

- No discounts (NetSellingPrice = MRP)
- Retail price is tax-inclusive
- Margin percentages are stable

5. Key Derived Metrics (Manufacturing)

Packs Per Quintal

PacksPerQuintal = 100000 / PackSize_Value

6. Total Retail Packs Produced

TotalRetailPacksProduced =

ProducedCapacityQuintalbyDay × PacksPerQuintal

Capacity Utilization %

ProducedCapacityQuintalbyDay / ProductionCapacityQuintalbyDay × 100

Utilization Classification

Range	Status
< 80%	Under Utilization
80–99%	Optimal
100%	Fully Utilized

7. Costing Logic

Total Operational Cost

PlantCost + MaintenanceCostbyday

Base Cost Per Pack (Produced)

BaseCostPerPack =

TotalOperationalCost / TotalRetailPacksProduced

Variants

- ProducedBaseCostPerPack
- ProductionBaseCostPerPack
- TargetBaseCostPerPack

These enable **scenario comparisons**.

8. Product Classification Logic

Main Product Category Derivation

Category	MainProductCategory
Beverages	Beverages
Snacks	Food
Breakfast	Food
Biscuits	Food

9. Pricing Ladder (Core of the Project)

Manufacturer Margin

Category	%
Beverages	25%
Food	20%

Distributor Margin

Category	%
Beverages	10%
Snacks	10%
Biscuits	7%
Breakfast	9%

Retailer Margin

Category	%
Beverages	20%
Snacks	22%
Biscuits	18%
Breakfast	15%

10. Pricing Formula Flow

BaseCostPerPack

→ ExFactoryPrice

→ WholesalePrice

→ PreTaxMRP

→ MRP

→ NetSellingPrice

→ ProfitPerPack

11. Tax Logic (India-Specific)

GST Era (2017 onwards)

Category	GST
Food	12%
Beverages	18%

Pre-GST (2015–2016)

Category	VAT
Food	5%
Beverages	12.5%

12. Profitability Metrics

Profit Per Pack

ProfitPerPack = NetSellingPrice – BaseCostPerPack

Profit Variants

- ProfitPerPack_Produced
- ProfitPerPack_Production
- ProfitPerPack_Expected

13. Advanced Analytics

Monthly Average Production

Using SQL Window Functions:

AVG(ProducedCapacityQuintalbyDay)

OVER (PARTITION BY Year, Month)

Variance Analysis

- Target vs Actual
- Installed vs Actual
- Cost & profit leakage

14. Final Output Object

Database Object

VW_FMCG_MANUFACTURING_PRICING_ANALYTICS

Contains

- Manufacturing KPIs
- Capacity utilization
- Pricing at all stages
- MRP & NetSellingPrice
- Profitability metrics
- Scenario comparison columns

15. BI & Reporting Use Cases

Power BI Dashboards

- Plant performance overview
- Cost & pricing waterfall
- Category profitability
- Utilization vs profit
- Historical tax comparison

Business Users

- Operations Head
- Finance Team
- Pricing Team
- Management